Mini Project

(Group Project | **2–3 students per group**)

I. Project Objective (What & Why)

The goal of this Mini Project is to:
reproduce and improve a classic Kaggle competition task, and go through a complete, standard deep learning modeling pipeline, including:

- Data understanding and problem formulation
- Proper validation strategy design
- Baseline construction
- Model improvement and hyperparameter tuning
- Result reproduction, analysis, and summary

You are expected to experience a **"real competition + academic-standard"** modeling process, rather than merely training a single model to chase leaderboard scores.

II. Project Format & Final Deliverables

Team Structure

- **2–3 students per group**
- Each group selects **one Kaggle competition**

Required Final Outcome

1. **Final Report (PDF)** (50%)
2. **Presentation (in-class)** (50%)

III. Optional Task List (Choose 1 out of 6)

Please select **one** Kaggle competition from the list below as your Mini Project topic.

> **Choosing a more difficult task yields a higher potential score ceiling during grading**, provided that the **basic requirements are met**:

> - **Introductory tasks**: standard score ceiling
> - **Intermediate tasks**: moderately higher ceiling
> - **Challenging tasks**: highest ceiling
>
> In other words: **higher difficulty + sufficient quality ⇒ greater scoring potential**.
>
> **Important note**: Difficulty weighting **does not lower requirements**.
> Even for challenging tasks, poor validation design or weak reproducibility can still result in low scores.

【Introductory】

1. Facial Keypoints Regression

- Link: https://www.kaggle.com/competitions/facial-keypoints-detection
- Task type: Image regression (multi-output)
- Characteristics:
  - Image preprocessing
  - Missing label handling
  - Mapping visual features to continuous numerical outputs

【Intermediate】

2. Mechanism of Action Prediction with High-Dimensional Biological Data

- Link: https://www.kaggle.com/competitions/lish-moa
- Task type: Multi-label classification
- Characteristics:
  - High-dimensional structured data
  - Strict cross-validation design
  - Feature engineering and regularization

3. Multimodal Radar Image Iceberg Classification

- Link: https://www.kaggle.com/competitions/statoil-iceberg-classifier-challenge
- Task type: Binary classification (images + numerical features)
- Characteristics:
  - Non-natural images

- Multimodal feature fusion
- Noise handling and robustness analysis

4. Automated English Writing Assessment in Educational Settings

- Link: https://www.kaggle.com/competitions/feedback-prize-english-language-learning
- Task type: Multi-target regression (NLP)
- Characteristics:
  - Long-text modeling
  - Multi-task learning
  - Generalization analysis

【Challenging】

5. Parkinson's Disease Progression Prediction from Time-Series Data

- Link: https://www.kaggle.com/competitions/amp-parkinsons-disease-progression-prediction
- Task type: Time-series regression
- Characteristics:
  - Temporal modeling
  - Medical structured data
  - Rigorous validation protocol

6. Large-Scale User Behavior Event Recommendation

- Link: https://www.kaggle.com/competitions/event-recommendation-engine-challenge
- Task type: Recommendation system / binary classification
- Characteristics:
  - High-dimensional sparse features
  - Multi-table joins
  - Cold-start problems

IV. Performance Target

Target Requirement: "Top-5-Level Performance"

Evaluation criteria depend on whether online Kaggle submission is still supported:

Case A: Competition Supports Late Submission (Online Scoring Available)

- Based on the **official Kaggle metric**
- Your result must:
  - Rank within the **Top 5** on the leaderboard

Case B: Competition Does NOT Support Late Submission

- A unified **5-fold cross-validation (5-fold CV)** score will be used
- Ranking within the **Top 5 based on CV results** is considered meeting the requirement
- If **direct scoring via Kaggle is not possible**, your Final Report must clearly specify:
  - Exact CV setup: number of folds, shuffling, random seeds
  - How evaluation metrics are computed (aligned with the competition metric)
  - How results can be reproduced (fixed seeds, environment description)

V. Academic Integrity & AI Usage Policy (Strictly Enforced)

Academic Integrity & Independence Disclosure

- **Prohibited behaviors**:
  - Directly copying or slightly modifying public Kaggle notebooks
  - Forking a notebook and only changing model names or parameters
- **Permitted behaviors (must be disclosed)**:
  - Referencing public ideas, papers, or discussion threads
  - Using public baselines as a starting point

AI Usage Policy

- **Permitted uses of AI tools**:
  1. Code debugging and error diagnosis
  2. Querying API or framework usage
  3. Idea discussion and technical explanation
  4. Language polishing (without altering technical content)
- **Explicitly prohibited behaviors**:
  1. Directly asking AI to **generate a complete, submission-ready solution and using it verbatim**

   2. Using AI to automatically reproduce or rewrite an entire Kaggle notebook
   3. Copying AI-generated implementation details without understanding them
- **Mandatory disclosure** (in the **“AI Tool Usage” section of the Final Report**):
   1. Which AI tools were used
   2. Their primary purposes (e.g., debugging, idea discussion)
   3. Which core designs were independently completed by the group

> **During the presentation, implementation details may be randomly questioned** to verify genuine understanding of your method.

VI. Model and Pretrained Model Usage Constraints

1. Pretrained Model Rules

- If the competition **explicitly allows or requires** pretrained models: follow the competition rules
- Otherwise:
   - **Pretrained model parameter count < 0.5B (500M)**

2. When Using Pretrained Models, You Must Specify

- Model source (checkpoint / paper / official link)
- Parameter count (official value or estimation method)
- Whether model ensembling is used

3. When Not Using Pretrained Models

- Clearly describe the model architecture
- Design motivation
- Improvements over the baseline

VII. Final Report Writing Guidelines

Important Notes

- **The Final Report must use the course-provided template** (to be released mid-semester)
- It is strongly recommended to prepare the report using the template from the beginning, rather than rushing at the end

The report must include at least the following core sections:

```
> # 1. Task Description & Metric
> Briefly introduce the competition background and motivation, describe the
modeling formulation (e.g., classification, regression, multi-label), and
clearly explain the official evaluation metric and its meaning.
>
> # 2. Data Analysis & Challenges
> Describe the dataset size, feature types, and basic distributions, and
analyze key challenges such as missing values, noise, imbalance, or inherent
modeling difficulties.
>
> # 3. Validation Strategy
> Explain the validation approach used (Kaggle leaderboard or cross-
validation), justify its suitability for the task, and detail specific
measures taken to prevent data leakage.
>
> # 4. Baseline Model
> Present a simple, reproducible baseline model, including basic modeling
ideas and training settings, along with its validation or leaderboard score.
>
> # 5. Model Improvements
> Systematically describe major improvements over the baseline (e.g., model
architecture, feature engineering, training strategies), and demonstrate
their effectiveness via comparative or ablation studies.
>
> # 6. Final Results
> Summarize the final model’s performance on the Kaggle leaderboard or via
cross-validation, and analyze how it compares to the “Top-5-level” target.
>
> # 7. AI Tool Usage Statement
> Clearly state whether and how AI tools were used (e.g., for debugging,
idea discussion, or writing assistance), and specify which core designs and
implementations were independently completed.
>
> # 8. Discussion & Reflection
> Reflect on which methods and attempts proved effective, which experiments
failed to meet expectations, and discuss current limitations and possible
future improvements.
>
> # 9. Reproducibility
> Provide key information required for reproducibility, including random
seed settings, runtime environment details, and access to complete code and
notebooks.
```

VIII. Important Reminders (Please Read Carefully)

- **Kaggle has a non-trivial learning curve** (accounts, notebooks, submissions, environments)
- Allocate sufficient time early in the project to overcome initial hurdles
- Do not wait until the final week to start running models